



IQ8P Microinverter

The extra high-powered, smart grid-ready Enphase IQ8P Microinverters are designed to match larger format commercial PV modules. The IQ8P has the highest energy production and reliability standards in the industry, and with rapid shutdown functionality, it meets the highest safety standards. The brain of the semiconductor-based microinverter is our proprietary, application-specific integrated circuit (ASIC) that enables the microinverter to operate in a grid-connected mode.



IQ Gateway

The IQ Gateway is the platform for energy management and integrates with the IQ Microinverters to provide complete control and insights into the Enphase Energy System.



Q-DCC-2-P adapter cable

Connect PV modules quickly and easily to IQ8P Microinverters using the included Q-DCC-2-P adapter cable with plug-and-play MC4 connectors.



IQ Relay three-phase

For production circuits in both single-phase and three-phase systems. IQ Relay acts as a grid monitoring and disconnection device and includes a built-in PLC phase coupler (three-phase).*



IQ Cabling

Install microinverters quickly and safely with IQ Cabling.

Compatible with the latest generation high-output PV modules

- Supports the latest high-current PV modules
- IQ8P product range supports all common PV module powers and cell architectures

Easy to install and commission

- Lightweight and compact
- Fast installation with simple AC cabling
- New integrated circuit technology enables faster firmware upgrades

High energy production, reliability, and safety

- More than one million power-on hours of reliability testing
- Patented Burst Mode technology provides increased energy production
- Low-voltage DC and rapid shutdown for the ultimate fire safety

Note:

Commissioning of IQ8P Microinverter systems requires Enphase Installer App version 3.31 or higher. IQ8P Microinverters cannot be mixed together with previous generations of Enphase microinverters (IQ7 Series, IQ6 Series) on the same IQ Gateway.

IQ8 Microinverters redefine reliability standards with more than one million cumulative hours of power-on testing, enabling an industry-leading limited warranty of 15 years.**

*IQ Relay is required to protect the PV system from grid abnormalities.

**15-year warranty is valid, provided an internet-connected IQ Gateway is installed. Get in touch with the Enphase team for warranty extension options.

IQ8P Microinverter

INPUT DATA (DC)		UNITS	IQ8P-72-2-INT
Typical module compatibility			60-cell/120-half-cell, 66-cell/132-half-cell, 72-cell/144-half-cell, 78-cell/156-half-cell No enforced DC/AC ratio and maximum input power. Modules can be paired as long as the maximum input voltage is not exceeded and the maximum input current of the inverter at the lowest and highest temperatures is respected. See the compatibility calculator at https://enphase.com/en-in/installers/micoinverters/calculator .
Minimum/Maximum input voltage	U_{dcmin}/U_{dcmax}	V	16/65
Start-up input voltage	$U_{dcstart}$	V	22
Rated input voltage	$U_{dc,r}$	V	45.5
Minimum/Maximum MPP voltage	U_{mppmin}/U_{mppmax}	V	36/55
Minimum/Maximum operating voltage	U_{opmin}/U_{opmax}	V	16/65
Maximum input current	I_{dcmax}	A	14
Maximum short-circuit DC input current	I_{scmax}	A	25 Maximum short-circuit current allowed for modules paired with IQ8P Microinverters: 20 A
Maximum input power ¹	P_{dcmax}	W	670
OUTPUT DATA (AC)		UNITS	IQ8P-72-2-INT
Maximum apparent power	$S_{ac,max}$	VA	480
Rated power	$P_{ac,r}$	W	475
Nominal grid voltage	U_{acnom}	V	230
Minimum/Maximum grid voltage	U_{acmin}/U_{acmax}	V	184/276
Maximum output current	I_{acmax}	A	2.07
Nominal frequency	f_{nom}	Hz	50
Minimum/Maximum frequency	f_{min}/f_{max}	Hz	47/55
Maximum units per single/ Three-phase 20 A circuit	16 A/ I_{acmax}		7 (L+N)/21 (3L+N) For IQ Cable with 12 AWG stranded conductors designed with NEC standard and using a 1.25 safety factor, 16 A per phase is calculated as the maximum current according to NEC requirements. Breaker selection should be decided based on "Circuit current < Breaker rated current < Cable current capacity".
Protective class (all ports)			II
Total harmonic distortion		%	<5
Power factor setting			1.0
Power factor range	$\cos \phi$		0.80 leading...0.80 lagging
Inverter maximum efficiency	η_{max}	%	97.34
IS/IEC 61683 efficiency	η_{EU}	%	97.00
Inverter topology			Isolated (HF Transformer)
Nighttime power loss		mW	100
MECHANICAL DATA			IQ8P-72-2-INT
Ambient air temperature range			-40°C to 65°C (-40°F to 149°F)
Relative humidity range			4% to 100% (condensing)
Overvoltage class AC port			III
Number of input DC connectors (pairs) per single MPP-tracker			1
AC connector type			IQ Cabling (refer to the individual datasheet for cable and accessories)
DC connector type			Supplied with Stäubli MC4 adapter
Dimensions (H x W x D)			265 mm (10.4") x 200 mm (7.9") x 35 mm (1.4") (without mounting brackets)
Weight (with mounting plate)			1.6 kg (3.5 lbs)
Cooling			Natural convection – no fans

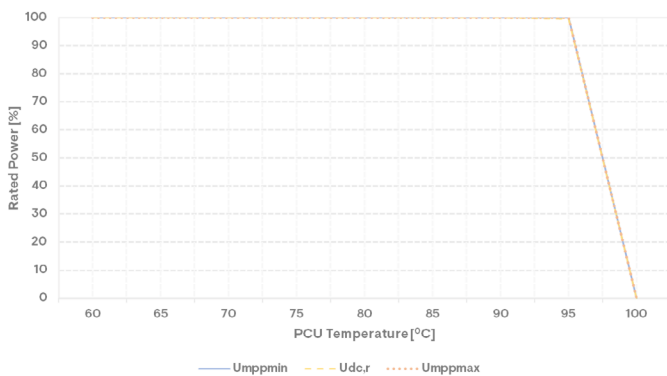
(1) Pairing PV modules with wattage above the limit may result in additional clipping losses. See the compatibility calculator at <https://enphase.com/en-in/installers/micoinverters/calculator>.

MECHANICAL DATA		IQ8P-72-2-INT	
Enclosure	Class II double-insulated, corrosion-resistant polymeric enclosure		
IP rating	Outdoor - IPX6/IP67		
Altitude	< 2,600 m		
Calorific value	59.25 MJ/unit		
STANDARDS		IQ8P-72-2-INT	
Grid compliance	IEC 61727		
Safety	EN IEC 62109-1, EN IEC 62109-2		
EMC	EN IEC 61000-3-2, 61000-3-3, 61000-6-2, 61000-6-3, EN IEC 50065-1, 50065-2-1, EN55011 ²		
Product labelling	CE, RCM, and BIS		
Advanced grid functions ³	Power export limiting (PEL), phase imbalance management (PIM), loss of phase detection (LOP), power factor control Q (U), cos (phi) (P)		
Microinverter communication	Power line communication (PLC) 110 – 120 kHz (Class B), narrowband 200 Hz		

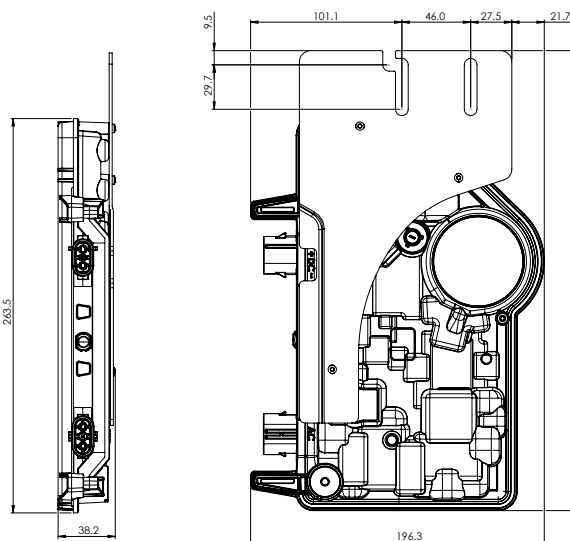
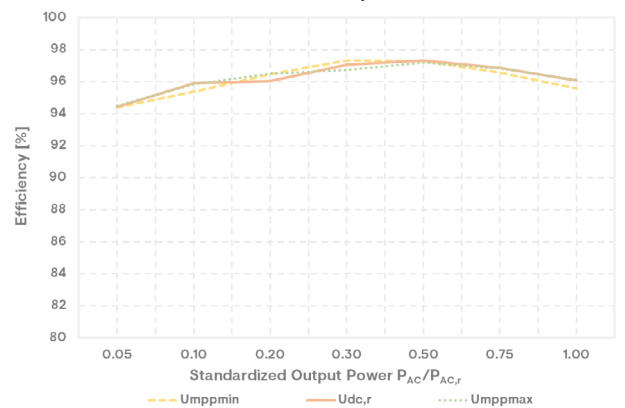
(2) At STC within MPP range.

(3) Some of these functions require IQ Gateway Metered with current transformers and/or IQ Relay installed.

IQ8P rated power v/s PCU temperature

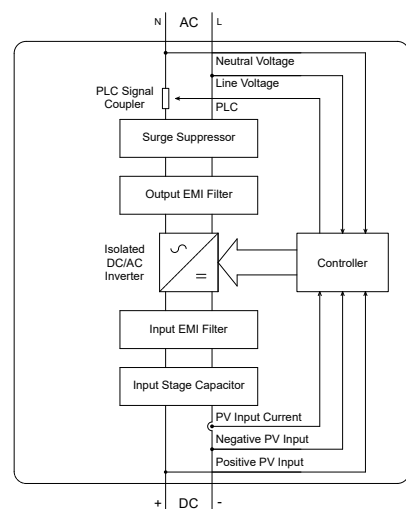


IQ8P efficiency curve



*All dimensions are in mm

Enphase IQ8P Microinverter



Assembled in India.

Manufacturer: Enphase Energy, Inc. 47281 Bayside Pkwy., Fremont, CA 94538, United States, PH: +1(707) 763-4784

Importer: Enphase Solar Energy Pvt. Ltd., IndiQube Golf View Homes, Ward No: 73 Airport, NAL Wind Tunnel Main Road, Bangalore-560017. Tel: +91-8061172500

Revision history

REVISION	DATE	DESCRIPTION
DSH-00055-2.0	September 2023	Updated Maximum short-circuit DC input current parameter to correctly reference to IQ8P
DSH-00055-1.0	August 2023	Preliminary release



भारतीय मानक ब्यूरो

(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)

BUREAU OF INDIAN STANDARDS

(Ministry of Consumer Affairs, Food & Public Distribution,
Govt. of India)

मानक भवन, 9 बहादुर शाह जफर मार्ग, नई दिल्ली - 110002

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi - 110002

दूरभाष/Phone: +91-11-23230856/2323010131/23233375/23239402

ई-मेल/E-mail: registration@bis.gov.in

वेबसाइट/Website: <https://bis.gov.in/>, <https://www.crsbis.in/BIS/>

Our Ref: REGISTRATION /CRS-2022-3292/R-61003093

Dated: 2024-07-19
00:21:48

RENEWAL ID : 27140

Subject : RENEWAL OF LICENCE R-61003093 AS PER IS 16221 (Part 2):2015/IEC 62109-2 :2011 & IS 16169 :2014/IEC 62116 :2008

Salcomp Manufacturing India Pvt Ltd.
NOKIA TELECOM SEZ, SIPCOT INDUSTRIAL PARK, PHASE III, SRIPERUMBUDUR 602105, INDIA.
TAMIL NADU, India, 602105



Dear Sir/Madam,

With reference to your online application dated 18-07-2024 for renewal of the above mentioned licence; this is to inform you that the same has been renewed from **22-09-2024** to **21-09-2029**.

It may be noted that the said licence granted under clause (b) of sub section (2) of section 13 of the Act shall *expire* at the end of the period for which it is granted unless renewed or its renewal is deferred. You are, therefore, requested to apply for next renewal to BIS within three months before the expiration of the licence.

Thanking you.

Yours faithfully,

Registration Department
Bureau of Indian Standards,
9, Bahadur Shah Zafar Marg,
New Delhi-110002.
Telfax : +91-11-23230856
E-mail: registration@bis.gov.in

Note: This is a system generated letter. Hence signature is not required.
To verify authentication of letter, kindly scan the QR code on this letter.

For details information on BIS, consult the e-BIS Portal (www.manakonline.in).
Please use BIS CARE APP for verification of ISI-marked goods and hallmarked gold jewellery.



Our Ref: REGISTRATION/CRS 2022-3292/R-61003093

Date:24-07-2023

Inclusion Id: 70946

Subject :Inclusion of Additional Model(s)

MANUFACTURING UNIT :	Salcomp Manufacturing India Pvt Ltd. NOKIA TELECOM SEZ, SIPCOT INDUSTRIAL PARK, PHASE III, SRIPERUMBUDUR 602105, INDIA. TAMIL NADU,India-602105 Jeganathan.VigneshPrakash@salcomp.com +919840692698	
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Dear Sir,

1. This has reference to your request for inclusion of models of "**Utility-Interconnected Photovoltaic Inverters**" as per **IS 16221 (Part 2):2015/IEC 62109-2 :2011 & IS 16169 :2014/IEC 62116 :2008 in Licence No. R-61003093** already granted to you which is valid upto 21-09-2024.

2. It is intimated that the additional Models as per details given below have been agreed to be included in your scope of Licence. **R-61003093 w.e.f. 24-07-2023:**

Product Category	Utility-Interconnected Photovoltaic Inverters
Product Name	Utility-Interconnected Photovoltaic Inverters
IS No.	IS 16221 (Part 2):2015/IEC 62109-2 :2011 & IS 16169 :2014/IEC 62116 :2008
Brand (As Declared by Manufacturer):	ENPHASE with e-logo
Inclusion of Additional Models (w.e.f. 24-07-2023)	IQ8P-72-2-INT
Factory Address	NOKIA TELECOM SEZ, SIPCOT INDUSTRIAL PARK, PHASE III, SRIPERUMBUDUR 602105, INDIA. TAMIL NADU,India-602105

3. Other terms and conditions of the licence shall remain same.

4. This letter is being issued with the approval of competent authority.

Kindly acknowledge receipt of this letter.

Thanking you,

Yours faithfully,
(Avik Datta)
Scientist-D
Telfax : +91-11-23230856
E-mail: registration@bis.gov.in

Note: This is a system generated letter. Hence signature is not required.
To verify authentication of letter, kindly scan the QR code on this letter.

For details information on BIS, consult the e-BIS Portal (www.manakonline.in).
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